KAIZEN MATHS

Lesson starter

Histograms and Frequency Density

Tables & Density

Name _____

SA1

Complete the missing values in the table for parcels.

Class interval	Class width	Frequency	Frequency density
$0 \leq x < 10$	10	10	?
$10 \leq x < 15$	5	?	2.50
$15 \leq x < 25$	10	35	?
$25 \leq x < 35$	?	30	3
$35 \leq x < 50$	15	?	2.50

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KAIZEN MATHS

Lesson starter

Histograms and Frequency Density

Tables & Density

Name _____

SB1

Complete the missing values in the table for plants.

Class interval	Class width	Frequency	Frequency density
$30 \leq x < 45$	15	45	?
$45 \leq x < 60$	15	?	2.50
$60 \leq x < 70$	10	25	?
$70 \leq x < 75$	?	5	1
$75 \leq x < 85$	10	?	1.50

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KAIZEN MATHS

Lesson starter

Histograms and Frequency Density

Tables & Density

Name _____

SC1

Complete the missing values in the table for candidates.

Class interval	Class width	Frequency	Frequency density
$40 \leq x < 60$	20	40	?
$60 \leq x < 80$	20	?	2.50
$80 \leq x < 95$	15	45	?
$95 \leq x < 105$	?	25	2.50
$105 \leq x < 115$	10	?	2.50

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KAIZEN MATHS

Lesson starter

Histograms and Frequency Density

Tables & Density

Name _____

SD1

Complete the missing values in the table for journeys.

Class interval	Class width	Frequency	Frequency density
$0 \leq x < 10$	10	10	?
$10 \leq x < 30$	20	?	1
$30 \leq x < 40$	10	20	?
$40 \leq x < 60$	?	30	1.50
$60 \leq x < 80$	20	?	3.50

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KAIZEN MATHS

Exit ticket

Histograms and Frequency Density

Tables & Density

Name _____

EA1

Complete the missing values in the table for students.

Class interval	Class width	Frequency	Frequency density
$0 \leq x < 15$	15	22.50	?
$15 \leq x < 35$	20	?	1
$35 \leq x < 45$	10	20	?
$45 \leq x < 60$	?	60	4
$60 \leq x < 80$	20	?	2.50

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KAIZEN MATHS

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Histograms and Frequency Density

Tables & Density

Name _____

EB1

Complete the missing values in the table for candidates.

Class interval	Class width	Frequency	Frequency density
$0 \leq x < 10$	10	35	?
$10 \leq x < 30$	20	?	2
$30 \leq x < 45$	15	15	?
$45 \leq x < 60$	?	37.50	2.50
$60 \leq x < 70$	10	?	1.50

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Histograms and Frequency Density

Tables & Density

Name _____

EC1

Complete the missing values in the table for candidates.

Class interval	Class width	Frequency	Frequency density
$40 \leq x < 50$	10	15	?
$50 \leq x < 55$	5	?	1
$55 \leq x < 65$	10	25	?
$65 \leq x < 75$	?	10	1
$75 \leq x < 80$	5	?	1.50

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Histograms and Frequency Density

Tables & Density

Name _____

EA1

Complete the missing values in the table for students.

Class interval	Class width	Frequency	Frequency density
$0 \leq x < 15$	15	22.50	?
$15 \leq x < 35$	20	?	1
$35 \leq x < 45$	10	20	?
$45 \leq x < 60$	?	60	4
$60 \leq x < 80$	20	?	2.50

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Teacher answers & working

Histograms and Frequency Density · Tables & Density

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SA1

Complete the missing values in the table for parcels.

Class interval	Class width	Frequency	Frequency density
$0 \leq x < 10$	10	10	?
$10 \leq x < 15$	5	?	2.50
$15 \leq x < 25$	10	35	?
$25 \leq x < 35$	?	30	3
$35 \leq x < 50$	15	?	2.50

Answer:

0 ≤ x < 10 density 1; 10 ≤ x < 15 frequency 12.50; 15 ≤ x < 25 density 3.50; 25 ≤ x < 35 width 10; 35 ≤ x < 50 frequency 37.50

Working:

Use the three connected facts:

class width = upper class boundary - lower class boundary

frequency density = frequency ÷ class width

frequency = class width × frequency density

For 0 ≤ x < 10, use frequency density = frequency ÷ class width.

frequency density = 10 ÷ 10 = 1

For 10 ≤ x < 15, use frequency = class width × frequency density.

frequency = 5 × 2.50 = 12.50

For 15 ≤ x < 25, use frequency density = frequency ÷ class width.

frequency density = 35 ÷ 10 = 3.50

For 25 ≤ x < 35, use class width = frequency ÷ frequency density.

class width = 30 ÷ 3 = 10

For 35 ≤ x < 50, use frequency = class width × frequency density.

frequency = 15 × 2.50 = 37.50

0 ≤ x < 10 density 1; 10 ≤ x < 15 frequency 12.50; 15 ≤ x < 25 density 3.50; 25 ≤ x < 35 width 10; 35 ≤ x < 50 frequency 37.50

Teacher answers & working

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SB1

Complete the missing values in the table for plants.

Class interval	Class width	Frequency	Frequency density
$30 \leq x < 45$	15	45	?
$45 \leq x < 60$	15	?	2.50
$60 \leq x < 70$	10	25	?
$70 \leq x < 75$	?	5	1
$75 \leq x < 85$	10	?	1.50

Answer:

30 ≤ x < 45 density 3; 45 ≤ x < 60 frequency 37.50; 60 ≤ x < 70 density 2.50; 70 ≤ x < 75 width 5; 75 ≤ x < 85 frequency 15

Working:

Use the three connected facts:

class width = upper class boundary - lower class boundary

frequency density = frequency ÷ class width

frequency = class width × frequency density

For 30 ≤ x < 45, use frequency density = frequency ÷ class width.

frequency density = 45 ÷ 15 = 3

For 45 ≤ x < 60, use frequency = class width × frequency density.

frequency = 15 × 2.50 = 37.50

For 60 ≤ x < 70, use frequency density = frequency ÷ class width.

frequency density = 25 ÷ 10 = 2.50

For 70 ≤ x < 75, use class width = frequency ÷ frequency density.

class width = 5 ÷ 1 = 5

For 75 ≤ x < 85, use frequency = class width × frequency density.

frequency = 10 × 1.50 = 15

30 ≤ x < 45 density 3; 45 ≤ x < 60 frequency 37.50; 60 ≤ x < 70 density 2.50; 70 ≤ x < 75 width 5; 75 ≤ x < 85 frequency 15

Teacher answers & working

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SC1

Complete the missing values in the table for candidates.

Class interval	Class width	Frequency	Frequency density
$40 \leq x < 60$	20	40	?
$60 \leq x < 80$	20	?	2.50
$80 \leq x < 95$	15	45	?
$95 \leq x < 105$	?	25	2.50
$105 \leq x < 115$	10	?	2.50

Answer:

40 ≤ x < 60 density 2; 60 ≤ x < 80 frequency 50; 80 ≤ x < 95 density 3; 95 ≤ x < 105 width 10; 105 ≤ x < 115 frequency 25

Working:

Use the three connected facts:

class width = upper class boundary - lower class boundary

frequency density = frequency ÷ class width

frequency = class width × frequency density

For 40 ≤ x < 60, use frequency density = frequency ÷ class width.

frequency density = 40 ÷ 20 = 2

For 60 ≤ x < 80, use frequency = class width × frequency density.

frequency = 20 × 2.50 = 50

For 80 ≤ x < 95, use frequency density = frequency ÷ class width.

frequency density = 45 ÷ 15 = 3

For 95 ≤ x < 105, use class width = frequency ÷ frequency density.

class width = 25 ÷ 2.50 = 10

For 105 ≤ x < 115, use frequency = class width × frequency density.

frequency = 10 × 2.50 = 25

40 ≤ x < 60 density 2; 60 ≤ x < 80 frequency 50; 80 ≤ x < 95 density 3; 95 ≤ x < 105 width 10; 105 ≤ x < 115 frequency 25

Teacher answers & working

Histograms and Frequency Density · Tables & Density

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SD1

Complete the missing values in the table for journeys.

Class interval	Class width	Frequency	Frequency density
$0 \leq x < 10$	10	10	?
$10 \leq x < 30$	20	?	1
$30 \leq x < 40$	10	20	?
$40 \leq x < 60$	?	30	1.50
$60 \leq x < 80$	20	?	3.50

Answer:

0 ≤ x < 10 density 1; 10 ≤ x < 30 frequency 20; 30 ≤ x < 40 density 2; 40 ≤ x < 60 width 20; 60 ≤ x < 80 frequency 70

Working:

Use the three connected facts:

class width = upper class boundary - lower class boundary

frequency density = frequency ÷ class width

frequency = class width × frequency density

For 0 ≤ x < 10, use frequency density = frequency ÷ class width.

frequency density = 10 ÷ 10 = 1

For 10 ≤ x < 30, use frequency = class width × frequency density.

frequency = 20 × 1 = 20

For 30 ≤ x < 40, use frequency density = frequency ÷ class width.

frequency density = 20 ÷ 10 = 2

For 40 ≤ x < 60, use class width = frequency ÷ frequency density.

class width = 30 ÷ 1.50 = 20

For 60 ≤ x < 80, use frequency = class width × frequency density.

frequency = 20 × 3.50 = 70

0 ≤ x < 10 density 1; 10 ≤ x < 30 frequency 20; 30 ≤ x < 40 density 2; 40 ≤ x < 60 width 20; 60 ≤ x < 80 frequency 70

Teacher answers & working

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EA1

Complete the missing values in the table for students.

Class interval	Class width	Frequency	Frequency density
$0 \leq x < 15$	15	22.50	?
$15 \leq x < 35$	20	?	1
$35 \leq x < 45$	10	20	?
$45 \leq x < 60$	?	60	4
$60 \leq x < 80$	20	?	2.50

Answer:

0 ≤ x < 15 density 1.50; 15 ≤ x < 35 frequency 20; 35 ≤ x < 45 density 2; 45 ≤ x < 60 width 15; 60 ≤ x < 80 frequency 50

Working:

Use the three connected facts:

class width = upper class boundary - lower class boundary

frequency density = frequency ÷ class width

frequency = class width × frequency density

For 0 ≤ x < 15, use frequency density = frequency ÷ class width.

frequency density = 22.50 ÷ 15 = 1.50

For 15 ≤ x < 35, use frequency = class width × frequency density.

frequency = 20 × 1 = 20

For 35 ≤ x < 45, use frequency density = frequency ÷ class width.

frequency density = 20 ÷ 10 = 2

For 45 ≤ x < 60, use class width = frequency ÷ frequency density.

class width = 60 ÷ 4 = 15

For 60 ≤ x < 80, use frequency = class width × frequency density.

frequency = 20 × 2.50 = 50

0 ≤ x < 15 density 1.50; 15 ≤ x < 35 frequency 20; 35 ≤ x < 45 density 2; 45 ≤ x < 60 width 15; 60 ≤ x < 80 frequency 50

Teacher answers & working

Histograms and Frequency Density · Tables & Density

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EB1

Complete the missing values in the table for candidates.

Class interval	Class width	Frequency	Frequency density
$0 \leq x < 10$	10	35	?
$10 \leq x < 30$	20	?	2
$30 \leq x < 45$	15	15	?
$45 \leq x < 60$	?	37.50	2.50
$60 \leq x < 70$	10	?	1.50

Answer:

0 ≤ x < 10 density 3.50; 10 ≤ x < 30 frequency 40; 30 ≤ x < 45 density 1; 45 ≤ x < 60 width 15; 60 ≤ x < 70 frequency 15

Working:

Use the three connected facts:

class width = upper class boundary - lower class boundary

frequency density = frequency ÷ class width

frequency = class width × frequency density

For 0 ≤ x < 10, use frequency density = frequency ÷ class width.

frequency density = 35 ÷ 10 = 3.50

For 10 ≤ x < 30, use frequency = class width × frequency density.

frequency = 20 × 2 = 40

For 30 ≤ x < 45, use frequency density = frequency ÷ class width.

frequency density = 15 ÷ 15 = 1

For 45 ≤ x < 60, use class width = frequency ÷ frequency density.

class width = 37.50 ÷ 2.50 = 15

For 60 ≤ x < 70, use frequency = class width × frequency density.

frequency = 10 × 1.50 = 15

0 ≤ x < 10 density 3.50; 10 ≤ x < 30 frequency 40; 30 ≤ x < 45 density 1; 45 ≤ x < 60 width 15; 60 ≤ x < 70 frequency 15

Teacher answers & working

Histograms and Frequency Density · Tables & Density

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EC1

Complete the missing values in the table for candidates.

Class interval	Class width	Frequency	Frequency density
$40 \leq x < 50$	10	15	?
$50 \leq x < 55$	5	?	1
$55 \leq x < 65$	10	25	?
$65 \leq x < 75$	?	10	1
$75 \leq x < 80$	5	?	1.50

Answer:

40 ≤ x < 50 density 1.50; 50 ≤ x < 55 frequency 5; 55 ≤ x < 65 density 2.50; 65 ≤ x < 75 width 10; 75 ≤ x < 80 frequency 7.50

Working:

Use the three connected facts:

class width = upper class boundary - lower class boundary

frequency density = frequency ÷ class width

frequency = class width × frequency density

For 40 ≤ x < 50, use frequency density = frequency ÷ class width.

frequency density = 15 ÷ 10 = 1.50

For 50 ≤ x < 55, use frequency = class width × frequency density.

frequency = 5 × 1 = 5

For 55 ≤ x < 65, use frequency density = frequency ÷ class width.

frequency density = 25 ÷ 10 = 2.50

For 65 ≤ x < 75, use class width = frequency ÷ frequency density.

class width = 10 ÷ 1 = 10

For 75 ≤ x < 80, use frequency = class width × frequency density.

frequency = 5 × 1.50 = 7.50

40 ≤ x < 50 density 1.50; 50 ≤ x < 55 frequency 5; 55 ≤ x < 65 density 2.50; 65 ≤ x < 75 width 10; 75 ≤ x < 80 frequency 7.50